# Chapter 15: Green Vehicle Routing

Vehicle Routing: Problems, Methods, and Applications (2nd ed., 2014)

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# Chapter Overview

# Why Does Routing Affect the Environment? I

Transport networks generate substantial economic growth but simultaneously impose significant negative externalities on society. The transport sector contributes roughly **13–14%** of global greenhouse-gas (GHG) emissions, making it one of the largest single-sector contributors. Road freight in particular is responsible for a disproportionate share because heavy goods vehicles are fuel-intensive and operate on diesel, which releases carbon dioxide ($CO_2$), nitrogen oxides ($NO_x$), particulate matter (PM), and other pollutants.

## The Core Question

Traditional Vehicle Routing Problems (VRP) seek to *minimise total distance driven* or *total fleet cost*. Green VRP asks a different question: can we design routes that *also* minimise fuel consumption and exhaust emissions — without abandoning the economic objectives entirely?

# Why Does Routing Affect the Environment? II

The rationale for studying green routing is threefold:

1. **Regulatory pressure.** Governments around the world have adopted mandatory emission targets (e.g. the European Union's Emissions Trading System, the International Road Transport Union targets). Carriers that exceed emission caps face financial penalties.
2. **Fuel cost savings.** Fuel typically constitutes 25–35% of total operating costs for a logistics company. Any reduction in fuel burn directly improves the bottom line.
3. **Corporate social responsibility.** Customers, investors, and regulators increasingly scrutinise the environmental footprint of supply chains. Demonstrating a commitment to green operations is a competitive advantage.

**Key Insight**

Minimising distance driven is *not* the same as minimising fuel consumption. A vehicle travelling a shorter route at high speed on a motorway may burn *more* fuel than one taking a slightly longer but flatter road at a steady, moderate speed. Speed, load, road gradient, and driving behaviour all matter.

# Why Does Routing Affect the Environment? III

## External Costs of Road Transport

The environmental damage caused by transport activities has long been recognised. The most prominent environmental costs of freight transport are:

- **Air pollution:** emissions of $CO_2$, $NO_x$, CO (carbon monoxide), $SO_x$ (sulphur oxides), and particulate matter affect human health and climate.
- **Noise:** vehicle noise causes stress and sleep disruption in urban populations.
- **Accidents:** road-freight vehicles are responsible for a disproportionate share of road fatalities.

According to international estimates, the external costs of road freight account for approximately **1–2%** of GDP in developed economies.

# Why We Need Accurate Fuel Models I

Before we can *optimise* emissions, we need to *measure* them. This section describes the mathematical models used to estimate how much fuel a vehicle burns — and hence how much $CO_2$ it emits — as a function of its operating conditions.

Two broad categories of emission models are used in the green VRP literature:

1. **Average-speed models:** estimate emissions as a function of the average speed over an entire trip or road segment. These are computationally simple and are the most commonly used in routing optimisation.
2. **Instantaneous (second-by-second) models:** estimate emissions at each moment in time, based on current engine power demand. These are highly accurate but require detailed second-by-second speed profiles and are computationally expensive.

# Why We Need Accurate Fuel Models II

## 15.2.1 Average Speed Models

The simplest and most widely used family of emission models estimates fuel consumption as a polynomial function of average speed $v$ (km/h). The general form proposed by the MEET project (Methodologies for Estimating and Reporting Emissions from Transport) for a given vehicle class is:

$$F(v) \;=\; K + \frac{a}{v} + b\,v + c\,v^2 + d\,v^3 + \frac{e}{v^2}$$

## Notation

- $F(v)$: fuel consumption in millilitres (ml) per kilometre at average speed $v$.
- $v$: vehicle speed in km/h.
- $K, a, b, c, d, e$: empirically calibrated coefficients that depend on the vehicle class (e.g. heavy goods vehicle over 32 tonnes vs. light van). These coefficients are provided in look-up tables by the MEET project.

# Why We Need Accurate Fuel Models III

## How to Read This Formula

The formula says that fuel consumption per kilometre is *not monotone* in speed:

- At **very low speeds** (e.g. stop-and-go traffic), the $a/v$ term dominates — idling and frequent acceleration waste fuel.
- At **very high speeds** (motorway), the $c\,v^2$ and $d\,v^3$ terms dominate — aerodynamic drag increases quadratically with speed.
- There is a **sweet spot** (often around 70–90 km/h for a heavy truck) where fuel per km is minimised.

This *convex* (U-shaped) relationship is the foundation of *eco-driving*: drive at the optimal speed to minimise fuel.

# Why We Need Accurate Fuel Models IV

# The Green Vehicle Routing Problem (GVRP) I

Now that we can *measure* fuel consumption on any arc, we can pose the optimisation problem: design a set of vehicle routes that minimises *total fuel consumption* (and therefore $CO_2$ emissions) while still serving every customer.
This problem is called the **Green Vehicle Routing Problem (GVRP)**. It extends the classical Capacitated VRP (CVRP) by replacing the distance-minimisation objective with a fuel-minimisation objective.

# The Green Vehicle Routing Problem (GVRP) II

# The Pollution Routing Problem (PRP) I

The **Pollution Routing Problem (PRP)**, introduced by Bektaș and Laporte (2011), extends the GVRP by making vehicle *speed a decision variable* on each arc. Rather than assuming a fixed speed everywhere, the PRP asks: what is the optimal speed on *each road segment* to minimise total fuel and emissions, subject to time-window constraints?

This is a much richer and more realistic model than the GVRP, because:

- Speed directly controls the fuel-consumption rate.
- Time windows create a tension: slower speed saves fuel but risks missing delivery windows.
- The optimal speed depends on arc length, cargo load, and the slackness of the time window at the next customer.

# The Pollution Routing Problem (PRP) II

# Algorithms for the GVRP and PRP I

Solving the GVRP and PRP in practice requires effective algorithms. The literature has proposed several approaches, ranging from exact mixed-integer programming to population-based metaheuristics.

## Tabu Search for the PRP (Bektas and Laporte, 2011)

Tabu search (TS) is a local-search metaheuristic that escapes local optima by *allowing moves that worsen the objective*, while maintaining a **tabu list** of recently visited solutions to prevent cycling. In the context of the PRP, TS works as follows:

**Why tabu search?** The PRP solution space is a combination of route structure (which customer is in which route and in what order) and speed choices on each arc. Standard descent methods get trapped in local optima because improving the route structure often requires temporarily making speed choices worse. By remembering and forbidding recently tried route moves (the "tabu list"), the algorithm can escape such traps.

## PRP Tabu Search Pseudocode

1. **Initialise:** generate a feasible solution $s_0$ using a nearest-neighbour heuristic. Set $s^* \leftarrow s_0$ (best solution), tabu list $T \leftarrow \emptyset$, iteration $k \leftarrow 0$.
2. **Optimise speeds:** for the current route structure, solve the inner speed-optimisation problem (a convex programme in the continuous speeds $v_{ij}$) to find optimal speeds. This can be done analytically arc by arc using calculus.
3. **Generate neighbourhood:** apply route-restructuring moves — relocate (move one customer to another route), 2-opt (reverse a segment), or swap (exchange two customers between routes).
4. **Select best non-tabu move:** choose the neighbourhood move with the best objective value that is *not* in the tabu list $T$. If the resulting solution is better than $s^*$, update $s^*$.
5. **Update tabu list:** add the selected move to $T$; remove the oldest entry if $|T|$ exceeds the tabu tenure $\tau$ (tau, a parameter, typically 10–20).
6. **Repeat** from step 2 until stopping criterion (maximum iterations or time limit) is met.
7. **Return** $s^*$.

# Algorithms for the GVRP and PRP III

## Key Insight: Separating Route and Speed Optimisation

A crucial algorithmic observation is that once the route structure is fixed, the speed-optimisation sub-problem can be solved *exactly and efficiently*:

For a single arc $(i, j)$ with distance $d_{ij}$, load $f_{ij}$, and time-window slack $\Delta_{ij}$, the optimal speed satisfies:

$$v_{ij}^* = \min\Big(\bar{v}_{\max},\ \max\Big(\bar{v}_{\min},\ \Big(\frac{c_f \lambda(\alpha_{ij} w + f_{ij}) + c_w}{2\, c_f \beta \lambda}\Big)^{1/3}\Big)\Big)$$

where $\bar{v}_{\min}$ and $\bar{v}_{\max}$ are the minimum and maximum allowed speeds (km/h).

This formula is derived by setting $dC/dv = 0$ for the per-arc cost function and clipping to the feasible speed range. The cube-root structure arises because aerodynamic drag is proportional to $v^2$ and the driver wage is proportional to $1/v$.

## The Non-Linear Programming Approach (Jabali et al.)

Jabali, Van Woensel, and de Kok (2012) present a similar approach for a single-vehicle route. They model fuel consumption as:

$$F_{\text{total}} = \sum_{(i,j)\in\text{route}} d_{ij}\left(p + q\, v_{ij}^2\right)$$

where $p$ captures load-dependent consumption and $q$ captures aerodynamic drag. Given a fixed route, minimising this over $v_{ij}$ subject to time-window constraints is a **convex programme** that can be solved optimally by Lagrangian relaxation or KKT (Karush-Kuhn-Tucker) conditions.

Their key finding: the optimal speed on an arc increases when the time window at the next stop is tight, and the optimal speed *decreases* as vehicle load increases (because the load coefficient $p$ amplifies the time-cost).

## The MINLP Approach of Figliozzi (2010)

Figliozzi models the VRP with time windows and minimises a weighted combination of travel time and fuel. The formulation is:

$$\text{Min} \sum_{(i,j)\in A} \left[ w_1\, t_{ij} + w_2\, e_{ij}(v_{ij}) \right] x_{ij}$$

where:

- $t_{ij} = d_{ij}/v_{ij}$: travel time on arc $(i, j)$.
- $e_{ij}(v_{ij})$: emissions on arc $(i, j)$ as a function of speed.
- $w_1, w_2$: weights reflecting the relative importance of time vs. emissions (set by the decision maker).

Results on 27-node benchmark instances show that optimising speed reduces emissions by an average of **9%** compared to fixing speed at the time-window-forcing level, at the cost of slightly longer travel time.

# Eco-Driving: Optimising Speed on a Given Route I

In the previous section we simultaneously optimised *which route to take* and *at what speed*. A simpler but practically important problem is: given a **fixed route** (the sequence of customers is fixed), find the optimal speed on each arc.

This is called **speed optimisation** or **eco-driving optimisation**. It is relevant in practice because:

- Many logistics companies cannot change routes easily (due to contractual obligations) but *can* give drivers guidance on optimal cruising speeds.
- It serves as the inner sub-problem in the tabu search for the PRP.
- It can be extended to consider *traffic-dependent speeds*.

# Eco-Driving: Optimising Speed on a Given Route II

## Formulation: Speed Optimisation on a Fixed Route

Given a route $r = (0, i_1, i_2, \ldots, i_m, 0)$ visiting $m$ customers, with arc distances $d_k$ (km) and time windows $[a_k, b_k]$ at each stop, choose speeds $v_k$ (km/h) to:

$$\text{Minimise} \quad \sum_{k=1}^{m+1} d_k \, F(v_k, f_k)$$

subject to:

$$\begin{aligned} a_k \leq T_k \leq b_k & \qquad \text{time window at stop } k \\ T_k = T_{k-1} + s_{k-1} + d_{k-1}/v_{k-1} & \qquad \text{arrival time recurrence} \\ \bar{v}_{\min} \leq v_k \leq \bar{v}_{\max} & \qquad \text{speed bounds} \end{aligned}$$

where $f_k$ is the vehicle load on arc $k$ (decreasing by $q_{i_k}$ after each delivery) and $T_k$ is the arrival time at stop $k$.

# Eco-Driving: Optimising Speed on a Given Route III

## Analytical Solution for the Unconstrained Case

When time windows are loose enough that speed constraints never bind, the optimal speed on arc $k$ is found by setting $\partial C/\partial v_k = 0$:

$$v_k^* = \left(\frac{\text{wage rate} \cdot d_k + \text{const}}{2\, c_f \beta\gamma\lambda\, d_k}\right)^{1/3}$$

Notice that $v_k^*$ is the **same** on all arcs (when load is ignored), because the cubic structure means the optimal speed balances the time-proportional driver wage against the speed-squared aerodynamic drag — and this balance is independent of distance.

However, when **load** is included, heavier arcs (early in the route) have a lower optimal speed because the load term increases the effective cost of travelling fast (the load must also be accelerated).

# Eco-Driving: Optimising Speed on a Given Route IV

## Constant-Speed Algorithms (Bektaş and Laporte)

Several algorithms have been proposed for the case where speed is constrained to take one of a finite set of discrete levels:

1. **Kara, Kara, and Yetis (2007):** introduced a load-dependent fuel model and showed it can be linearised; proposed an integer programming formulation solved with a branch-and-bound algorithm.
2. **Xiao et al. (2012):** proposed a fuel consumption optimisation model for the CVRP, solved with a savings-based heuristic where savings are computed using the load-dependent fuel model rather than distance.
3. **Demir, Bektaş, and Laporte (2012):** proposed a large neighbourhood search (LNS) for the PRP, with destroy operators (random removal of a subset of customers) and repair operators (best-insertion with simultaneous speed optimisation).

# Eco-Driving: Optimising Speed on a Given Route V

## Key Finding: Savings from Speed Optimisation

Numerical experiments (Demir et al. 2012) on benchmark VRP instances with 100–200 customers show:

- Speed optimisation alone (fixing routes, optimising speeds) reduces fuel/emissions by **10–15%** on average compared to driving at a fixed default speed.
- Jointly optimising routes and speeds yields **18–25%** emission reductions compared to the traditional distance-minimising VRP.
- The benefit of speed optimisation is greatest when time-window slack is high: tightly constrained routes leave little room for eco-driving.

# Balancing Cost and Emissions: The Pareto Front I

In practice, fleet operators must balance two conflicting objectives:

1. **Economic cost** — vehicle operating costs, driver wages, fuel expenditure, fixed vehicle costs.
2. **Environmental impact** — $CO_2$ emissions, $NO_x$, particulate matter.

No single route simultaneously minimises both: a route that uses more vehicles (higher cost) may deliver faster and burn less fuel per unit, while a cost-efficient route might consolidate loads onto fewer vehicles and push speeds up to meet tight time windows.

# Balancing Cost and Emissions: The Pareto Front II

# Beyond Road Transport: Rail, Air, and Multimodal I

While most green VRP research focuses on road transport, significant opportunities for emission reduction also exist in **rail freight**, **air cargo**, and **multimodal transport**.

# Beyond Road Transport: Rail, Air, and Multimodal II

# Electric Vehicle Routing: The E-VRP I

The ultimate green routing scenario replaces fossil-fuel vehicles with **electric vehicles (EVs)**. EVs produce zero tailpipe emissions, and if the electricity grid is powered by renewables, the entire supply chain becomes carbon-neutral.

However, EVs introduce a new operational constraint: **limited range**. A standard electric delivery van has a range of roughly 100–200 km on a single charge (improving rapidly with battery technology). If the route requires more than this, the vehicle must stop at a **recharging station**.

This gives rise to the **Electric Vehicle Routing Problem (E-VRP)**, also described as "routing with range anxiety".

# Electric Vehicle Routing: The E-VRP II

# Conclusions: What Green VRP Has Achieved I

Chapter 15 has surveyed the rapidly growing field of **green vehicle routing**, covering the full spectrum from emission models to exact algorithms, from speed optimisation to electric vehicle routing.

# Conclusions: What Green VRP Has Achieved II

## Summary of Main Contributions

1. **Emission models.** The chapter reviewed average-speed models (MEET), instantaneous models (CMEM), and the load-speed decomposition model that underpins the GVRP and PRP. These models reveal that *speed and load are the primary drivers of fuel consumption* — not just distance.
2. **Green VRP and Pollution Routing Problem.** The PRP is the most comprehensive formulation, jointly optimising routes and speeds. Exact and heuristic solution methods achieve 18–25% emission reductions compared to traditional VRP.
3. **Speed optimisation.** Even on fixed routes, optimising vehicle speed across arcs reduces emissions by 10–15%.
4. **Multicriteria methods.** Pareto-front analysis reveals that modest cost increases (5–10%) can achieve substantial emission reductions (20–30%).
5. **Electric VRP.** The E-VRP framework handles range constraints and recharging station decisions. It is the most complex variant but represents the future of zero-emission logistics.

# Conclusions: What Green VRP Has Achieved III

## Future Research Directions

The chapter identifies several open problems and promising directions:

1. **Real-world traffic data integration.** Most models use simplified speed profiles. Incorporating real-time traffic data (GPS, road sensors) into routing decisions would make models more accurate and actionable.
2. **Dynamic green VRP.** Routes are currently planned in advance. Dynamic variants that re-optimise routes in real time as conditions change (traffic, new orders) are largely unexplored.
3. **Lifecycle emission assessment.** Current models focus on tailpipe emissions. A full lifecycle analysis would include the emissions from vehicle manufacturing, fuel production (well-to-wheel), and end-of-life recycling.
4. **Stochastic demand and emissions.** Travel speeds, fuel consumption, and customer demand are all uncertain in practice. Robust and stochastic programming approaches to green VRP remain an open research frontier.
5. **Fleet mix optimisation.** Most models assume a homogeneous fleet. Optimising the *composition* of a fleet (how many EVs, CNG trucks, and diesel vehicles) simultaneously with routing is a complex but important problem.
6. **Collaborative logistics.** Multiple carriers sharing vehicles and routes can dramatically reduce

# Python: GVRP Fuel Objective Implementation

Listing 1: Computing fuel consumption for a GVRP route using the load-dependent model.

```python
import numpy as np

def fuel_consumption_route(route, demands, dist, alpha=0.3, beta=0.01):
    """
    Compute total fuel for one route using the load-dependent model:
        F = sum_arcs [ alpha * d_ij * x_ij  +  beta * d_ij * f_ij ]

    Parameters
    ----------
    route   : list of node indices, e.g. [0, 3, 1, 5, 0]
              (must start and end at depot = 0)
    demands : list/array of customer demands (index 0 = depot demand = 0)
    dist    : 2-D distance matrix  dist[i][j] in km
    alpha   : empty-vehicle fuel rate (kg fuel / km)
    beta    : load-dependent fuel rate (kg fuel / km / kg cargo)

    Returns
    -------
    total_fuel : float (kg of fuel consumed)
    """
    # Compute load on each arc (decreasing as deliveries are made)
    total_fuel = 0.0
    load = sum(demands[c] for c in route[1:-1])   # start fully loaded

    for k in range(len(route) - 1):
```

# Python: Optimal Speed on a Single Arc

Listing 2: Compute the fuel-optimal speed on a single arc with time-window constraints.

```python
def optimal_speed_arc(d_ij, load, slack, c_f=1.2, c_w=5.0,
                      beta=0.0001, alpha_ij=1.2, v_min=50, v_max=110):
    """
    Optimal speed on arc (i->j) minimising fuel + driver wage cost.

    Fuel model (simplified):  F(v) = (alpha_ij + beta * v^2) * d_ij  litres
    Driver wage:              W(v) = c_w * d_ij / v
    Total cost:               C(v) = c_f * F(v) + W(v)

    Optimal (unconstrained):  v* = (c_w / (2 * c_f * beta))^(1/3)
    Then clip to [v_min, d_ij/slack] and [v_min, v_max].

    Parameters
    ----------
    d_ij     : arc distance (km)
    load     : vehicle load on this arc (tonnes)
    slack    : available time window slack on this arc (hours)
    c_f      : fuel cost (€/litre)
    c_w      : driver wage (€/hour)
    beta     : aerodynamic drag coefficient
    alpha_ij: constant fuel term (litres/km)
    v_min, v_max : speed bounds (km/h)

    Returns
    -------
```

# Python: Pareto Front Generation (Weighted-Sum Method)

Listing 3: Generate a Pareto front for cost vs. emissions using a weighted-sum scalarisation.

```python
import numpy as np
import matplotlib
matplotlib.use('Agg')
import matplotlib.pyplot as plt

def solve_weighted_vrp(w_cost, w_emission, instance):
    """
    Placeholder: solve a VRP minimising  w_cost*C + w_emission*E.
    In practice, call a tabu-search or exact solver here.
    Returns (cost, emission) for the best solution found.
    """
    # Simulated trade-off (replace with real solver output)
    # As w_emission increases, solutions become greener but costlier
    base_cost, base_emission = 2500, 100
    cost = base_cost * (0.8 + 0.4 * w_cost / (w_cost + w_emission + 1e-9))
    emission = base_emission * (0.3 + 0.7 * w_emission / (w_cost + w_emission + 1e-9))
    noise = np.random.randn() * 20
    return cost + noise, emission

def generate_pareto_front(instance, n_weights=15):
    """
    Sweep weight pairs to approximate the Pareto front.
    w_cost ranges from 0 to 1; w_emission = 1 - w_cost.
    """
    results = []
```

# Key References I

The following are the most important references cited in Chapter 15. They are listed here to help the reader locate primary sources for deeper study.


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# Chapter 15 Summary

## Topics Covered

- External costs of road freight
- Average-speed & instantaneous emission models (MEET, CMEM)
- Load-speed-fuel relationship
- Green VRP (GVRP) formulation
- Pollution Routing Problem (PRP)
- Speed optimisation on fixed routes
- Time-dependent VRP
- Multicriteria analysis: Pareto fronts
- Modal comparison (road/rail/air)
- Electric VRP (E-VRP) and range constraints
- Alternative fuels: CNG, hybrid, hydrogen

## Key Numbers to Remember

- Transport $\approx$ 13–14% of global GHG emissions
- Fuel $\approx$ 25–35% of logistics operating cost
- Optimal truck speed $\approx$ 70–90 km/h (fuel)
- Speed optimisation: $\sim$10–15% fuel savings
- Route + speed optimisation: $\sim$18–25% savings
- Modal shift (road $\rightarrow$ rail): $\sim$50–70% savings
- EV: up to 65% savings (clean electricity)
- Pareto: 5–10% cost increase buys 20–30% emission cut

## Open Problems